

GP2 Desk Research Update Roadmap

Review of the current GP2 desk research baseline

The current GP2 report situates documentation continuity within established reproducibility and lifecycle scholarship (e.g., FAIR principles, provenance, lifecycle models) and anchors “Institutional Benchmarking” in a set of governance and infrastructure references that are primarily either undated web pages or policy documents from 2021. [filecite\turn0file1](#)

In the “Institutional Benchmarking” paragraph of Section 1, the benchmark layer is constructed mainly from:

- a national funder requirement statement based on NWO ¹ (cited as n.d. in the report),
- international governance sources (OECD ² 2021; Science Europe ³ 2021),
- university RDM policy pages (e.g., University of Twente, n.d.),
- infrastructure providers (SURF ⁴, n.d.; 4TU.ResearchData, n.d.),
- and a local verification initiative (TU Delft Digital Competence Centre, n.d.). [filecite\turn0file1](#)

Appendix C makes clear that desk research was deliberately grouped and used for conceptual grounding, methodological grounding, and framework derivation, with a retention principle that sources were only kept if they directly supported the diagnosis, method rigor, or solution design. [filecite\turn0file0](#)

Appendix D documents how your theoretical model selection was staged and traceable (desk research → model selection → operationalisation), reinforcing that the problem is not “missing research process,” but “missing benchmark currency and trend coverage.” [filecite\turn0file0](#)

Inventory of the existing source base and what it already covers

A review of the report’s reference list shows a source base that is strong on foundational and methodological legitimacy but thin on *2023–2026 trend evidence*. The reference list contains many undated institutional pages and foundational works, and only a small number of recent (post-2022) publications. [filecite\turn0file1](#)

A simple way to show why this reads “dated” to an assessor: in the GP2 reference list, the most recent dated items are limited (e.g., a single 2023 article and a single 2025 article), with no 2024 benchmark sources visible, while the benchmark paragraph itself relies primarily on 2021 governance and n.d. institutional sources. [filecite\turn0file1](#)

What your existing benchmark sources already substantiate well

Your existing benchmark layer does substantiate several important baseline claims:

- **Documentation is structurally expected in modern research governance.** This is supported by OECD and Science Europe guidance already cited in the report. [\[filecite\turn0file1\]](#)
- **Lifecycle approaches and FAIR-aligned expectations matter for reuse and continuity.** This is already supported through FAIR and lifecycle scholarship in your source base. [\[filecite\turn0file1\]](#)
- **Institutional infrastructure exists to support preservation and reuse.** This is already supported by infrastructure references (SURF, 4TU.ResearchData) and university policy pages. [\[filecite\turn0file1\]](#)

What your existing source base does not yet substantiate

Your current sources do *not* yet provide strong, explicit support for:

- **What has changed recently (2023–2026)** in open science governance *implementation*, monitoring, and incentives (as opposed to high-level principles).
- **How AI is already reshaping documentation, metadata generation, retrieval, and knowledge reuse** (the assessor’s explicit example).
- **Current user-side research on retrieval, trust, and reuse behaviors**, which would strengthen the stakeholder perspective and the “how do they search?” dimension. [\[filecite\turn0file1\]](#)

Why the assessor concluded “missing current data and trends”

Aline’s conclusion follows logically from what is *visible in the benchmark paragraph* and its citations:

- The benchmark paragraph is built around 2021 policy sources and n.d. institutional pages rather than 2023–2026 developments. [\[filecite\turn0file1\]](#)
- The benchmark does not discuss AI at all, and your current bibliography does not include any dedicated sources addressing generative AI in research governance or AI-driven documentation/knowledge management. [\[filecite\turn0file1\]](#)
- Recent developments that have become central in RDM and open science practice—such as stronger monitoring/assessment of open practices, machine-actionable workflows, evolving metadata standards, and AI-assisted discovery—are not explicitly benchmarked.

This is why Aline can accept the *execution* (“data collection according to plan”) yet still judge the *benchmark research layer* as dated and missing contemporary forces. [\[filecite\turn0file1\]](#)

Desk research plan for a 2023–2026 benchmark update

This plan is designed to produce “current data and trends” in a defensible way, while ensuring that added sources are genuinely new relative to the GP2 report and Appendix C.

Desk research questions that directly address the feedback

The update should be framed as a “supplementary benchmarking scan” answering four applied questions:

- **What are the most salient 2023–2026 policy and infrastructure shifts** affecting research documentation and RDM practice (especially in Europe and the Netherlands)?
- **How is AI already shaping documentation and knowledge management**, specifically through structuring, retrieval, summarisation, metadata generation, and reuse?
- **What are the current technical standards and infrastructure components** (metadata schemas, PIDs, packaging/provenance mechanisms) that enable continuity and reuse in practice today?
- **What does current empirical research say about user retrieval and reuse behaviors** (how reusers search, assess reusability, and build trust)?

Source types that should be prioritized

To satisfy the assessor’s expectation of “current trends,” the update should prioritize:

- **Recognised research governance bodies and funders** (EU, national funders, open science programs).
- **Infrastructure and standards organisations** (PID/metadata standards bodies; open science cloud/infrastructure programs).
- **Peer-reviewed empirical studies and systematic reviews** on AI impacts in KM and metadata/documentation systems.
- **Peer-reviewed user studies** on discovery, reuse, and trust in repositories.

Credibility and “newness” checks

For each candidate source, apply two filters:

- **Credibility filter:** policy source from a recognised body or peer-reviewed academic publication.
- **Newness filter:** not present in the GP2 reference list and not present in Appendix C’s desk research mapping.

The “newness filter” is especially important here: your GP2 already contains many foundational sources; the update must add what is *currently missing*, not add duplicates.

Curated new sources (2023–2026) and how each will be used in Section 1

The sources below are **not present** in the current GP2 reference list (and therefore qualify as genuinely new benchmark additions). Each source is mapped to a concrete sentence/argument you can insert into Section 1.

Mapping table for Section 1 insertions

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
Current “open science is operationalised and assessed” trend (not just policy ideals)	Science Europe ³ (2024) survey report on strategic approaches and research assessment of open science. ⁵	Open science is increasingly embedded in organisational strategy, monitoring, and research assessment, making documentation and reuse expectations more operational and measurable, not only aspirational. ⁵	Section 1.1 Institutional Benchmarking: after the sentence citing OECD/Science Europe 2021, add a “2024 survey evidence” sentence.
Netherlands-specific currency: funder expectations are actively changing	NWO ¹ (2025) updated RDM expectations page; NWO (2026) policy change about DMP template approval. ⁶	Shows that Dutch RDM expectations are actively evolving and shifting responsibilities to institutions (e.g., changes in DMP template approval). This directly delivers “current developments” rather than undated requirements. ⁶	Section 1.1 Institutional Benchmarking: replace or supplement “NWO, n.d.” with dated 2025/2026 citations and 1–2 sentences describing the shift.
Netherlands open science infrastructure investment trend	Open Science NL Work Programme 2024–2025 (published via OECD/Zenodo mirrors). ⁷	Demonstrates active national investment in open science support capacity, including roles and training for research data professionals, reinforcing that continuity is being institutionalised. ⁷	Section 1.1 Institutional Benchmarking: add a Dutch “infrastructure and professionalisation” sentence immediately after the SURF/4TU sentence.

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
AI is now explicitly part of the research governance context	European Commission ⁸ (2024) living/responsible use guidelines for generative AI in research. ⁹	Establishes that generative AI is now a governance concern requiring responsible use, transparency, and documentation of AI involvement in research workflows. ⁹	Section 1.1 Institutional Benchmarking: add a short paragraph titled (or signposted as) “AI as an emerging documentation driver,” explicitly tied to documentation and traceability requirements.
AI is reshaping education-and-research environments (relevant to student lab context)	UNESCO ¹⁰ (2023) guidance for generative AI in education and research. ¹¹	Provides global context: AI use in education/research is widespread enough to require guidance on governance, human oversight, and responsible integration—relevant because your setting is an applied educational research lab. ¹¹	Section 1.1 (AI paragraph) or Section 1 context setting: 1 sentence linking the VCH Lab context to broader higher-education AI guidance.
AI is directly transforming knowledge management practice	Rezaei (2025) “Artificial intelligence in knowledge management” (Technological Forecasting & Social Change). ¹²	Demonstrates that AI affects KM processes and introduces technological, organisational, and ethical challenges. Use it to justify why documentation/KM benchmarks must now treat AI as a structural force. ¹²	Section 1.1 AI paragraph: add a sentence linking AI-driven KM transformation to documentation, storage, retrieval, and privacy concerns.

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
AI automates metadata generation and affects accessibility and usability	Yang (2025) “Impact of modern AI in metadata management” (Springer). ¹³	Provides direct evidence that AI is being used for automated metadata generation, governance, and improved discovery/ access—exactly aligned to Aline’s “AI shaping documentation and management” example. ¹³	Section 1.1 AI paragraph: add 1–2 sentences on automated metadata generation as a trend and why inconsistent documentation makes automation harder.
AI changes what “research transparency” requires (AI disclosure is now part of provenance)	Jones (2025) “Generative AI in qualitative research... heuristic for disclosing uses of AI.” ¹⁴	Supports the claim that AI creates new transparency/ disclosure requirements, meaning documentation must increasingly include records of AI involvement, prompts, and validation decisions. ¹⁴	Section 1.3 (ethics/ credibility) or Section 4 (AI disclosure): add as evidence that AI disclosure is an emerging norm, not an optional add-on.
Research integrity bodies now require explicit AI acknowledgement practices	Committee on Publication Ethics ¹⁵ (2023) authorship and AI tools guidance; International Committee of Medical Journal Editors ¹⁶ (2024) updated recommendations on acknowledging AI assistance. ¹⁷	Establishes credible cross-sector expectation: AI tools are not “authors,” and human responsibility + appropriate acknowledgement is required. Useful to justify why documentation systems must explicitly record tool use and human decisions. ¹⁷	Section 1.3 ethics: 1 short sentence noting that AI involvement must be documented/ acknowledged per publication ethics guidance.

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
Technical trend: metadata standards are evolving to support richer, more machine-usable descriptions	DataCite ¹⁸ Metadata Working Group (2024) Metadata Schema 4.5. ¹⁹	Shows that metadata standards are actively evolving to support richer and more accurate metadata—useful to argue that inconsistent file structures and missing context reduce compatibility with modern metadata practices. ¹⁹	Section 1.1 benchmarking: add after “metadata standards and persistent identifiers” a sentence noting metadata standards are evolving (DataCite 4.5).
Infrastructure trend: Europe operationalises PIDs and minimum metadata across federated systems	EOSC Partnership Monitoring Framework (2024 draft). ²⁰	Provides evidence that European infrastructure programs set measurable targets for PID adoption and minimum metadata frameworks—strong “current trend” reinforcement beyond 2021 principles. ²⁰	Section 1.1 benchmarking: insert 1 sentence showing that cross-infrastructure discoverability requires minimum metadata + PID practices.
Packaging/ provenance trend: research objects are being standardised for reuse and traceability	RO-Crate 1.2 specification (release activity 2025; spec pages). ²¹	Substantiates that modern practice increasingly packages files + structured metadata together so projects are self-describing and reusable—directly relevant to your “file structures vary” and “reuse potential” objectives. ²¹	Section 1.1 or transition to Section 2: add a sentence that modern reuse approaches package outputs with machine-readable metadata (RO-Crate).

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
Current development: machine-actionable documentation is being pulled into “FAIR digital object” ecosystems	Miksa (2024) “Turning machine-actionable DMPs into FAIR digital objects.” ²²	Supports the argument that documentation is shifting from static narrative reports into machine-actionable, interoperable documentation components linked to services and infrastructure. ²²	Section 1.1 benchmarking: add a sentence about machine-actionable documentation as a trend (beyond PDFs and folders).
User-side empirical evidence: reuse depends on user strategies and reusability assessments	Jiang, Li, & Liu (2025) data reuse interviews (JASIST; summarized on RePEc). ²³	Provides empirical, interview-based evidence about reusers’ strategies, criteria, and challenges when reusing data—directly aligned with “how do they search” and “what gaps do they find.” ²³	Section 1.4 user perspective: add a short paragraph bridging from your interviews to external literature on reuse assessment and search strategies.
User-side empirical evidence: trust and decision-making in repositories	Yakel, Faniel, & Robert (2024) “data reuser trust in a digital repository” (project/publisher summaries). ²⁴	Supports the claim that usability, trust signals, and repository factors influence whether reusers will rely on materials—useful for framing why documentation must be interpretable and verifiable. ²⁴	Section 1.4 user perspective or criteria bridge: add 1 sentence on trust as a condition of reuse.

Trend / gap to fix	New source (2023–2026)	What it substantiates (use in Section 1)	Exact Section 1 insertion target
User-centric trend: discovery and reuse now recognizes “machine users” and AI-driven discovery workflows	Wu et al. (2026) user-centric recommendations for research data repositories (Data Science Journal). ²⁵	Shows current thinking that (1) FAIR is necessary but not sufficient for discovery usability, and (2) discovery ecosystems now involve both humans and automated/AI agents (“machine users”), raising the bar for structured metadata and clarity. ²⁵	Section 1.4 user perspective and Section 1.1 AI paragraph : connect user search behavior to AI-enabled discovery and metadata quality.
High-level synthesis for stakeholder perspective: reuse is knowledge exchange, not a simple file transfer	Borgman & colleagues (2025) “Distance matters” framing (HDSR). ²⁶	Provides a strong conceptual bridge: reuse fails when distance between creators and reusers is too large, and good curation/provenance shrinks that distance—highly compatible with your continuity diagnosis. ²⁶	Section 1.4 : use as a synthesis anchor to justify why “user perspective” is central to the solution criteria.

Bibliography and citation update guidance for the GP2 report

The goal here is not to delete your existing foundational sources (FAIR, provenance, lifecycle models, design science), but to **layer in dated 2023–2026 evidence** that makes the benchmark read current and trend-aware.

Replace or upgrade undated benchmark citations where possible

Your current benchmark cites “NWO, n.d.” as evidence of funder expectations. The updated NWO RDM page (2025) and the NWO change to institutional DMP template approval (effective March 2026) allow you to convert this into a current, date-stamped benchmark statement. ⁶

Similarly, adding Science Europe’s 2024 survey report shifts the benchmark from “principles exist” to “principles are operationalised through monitoring and assessment.” ⁵

Add a dedicated AI benchmark subsection

Your benchmark currently has no AI dimension. Aline's feedback explicitly expects this to be included. The EC's 2024 guidelines plus UNESCO's 2023 guidance provide the governance foundation, while Rezaei 2025 and Yang 2025 provide peer-reviewed evidence of AI transforming KM and metadata practice. ²⁷

Integrate user-side literature to support "who uses the materials"

Your interviews already document reuse difficulty, missing decisions, and fragmented storage, but adding current reuse/discovery literature makes the stakeholder perspective academically grounded and visibly current. Relevant additions include Borgman 2025, Jiang 2025, Yakel et al. 2024, and Wu et al. 2026. ²⁸

Copy-ready APA entries for the new sources

Science Europe. (2024). Strategic approaches to, and research assessment of, open science: Survey report. Science Europe.

European Commission, Directorate-General for Research and Innovation. (2024). Responsible use of generative AI in research: Guidelines developed by the European Research Area Forum.

UNESCO. (2023). Guidance for generative AI in education and research.

SURF. (2026). SURF tech trends 2026.

NWO. (2025). Research data management. Dutch Research Council (NWO).

NWO. (2026). Institutional data management plan templates no longer need to be approved by NWO and ZonMw. Dutch Research Council (NWO).

Open Science NL. (2024). Work programme 2024–2025. Open Science NL.

DataCite Metadata Working Group. (2024). DataCite metadata schema documentation for the publication and citation of research data and other research outputs (Version 4.5). DataCite e.V. doi:10.14454/g8e5-6293

EOSC Association / EOSC Partnership. (2024). EOSC partnership monitoring framework (draft).

Rezaei, M. (2025). Artificial intelligence in knowledge management: Identifying and addressing the key implementation challenges. Technological Forecasting and Social Change. doi:10.1016/j.techfore.2025.124183

Yang, W. (2025). The impact of modern AI in metadata management. Human-Centric Intelligent Systems.

Jones, K. M. L. (2025). Generative AI in qualitative research and related

transparency problems: A novel heuristic for disclosing uses of AI.
International Journal of Qualitative Methods. doi:10.1177/16094069251404329

Borgman, C. L., & colleagues. (2025). From data creator to data reuser: Distance matters. Harvard Data Science Review.

Wu, M., et al. (2026). Bridging the data discovery gap: User-centric recommendations for research data repositories. Data Science Journal. doi:10.5334/dsj-2026-006

Jiang, T., Li, W., & Liu, J. (2025). The landscape of data reuse in interactive information retrieval: Motivations, sources, and evaluation of reusability. Journal of the Association for Information Science and Technology. doi:10.1002/asi.25020

Yakel, E., Faniel, I. M., & Robert Jr., L. P. (2024). An empirical examination of data reuser trust in a digital repository. Journal of the Association for Information Science and Technology. doi:10.1002/asi.24933

Miksa, T. (2024). Turning machine-actionable data management plans into FAIR digital objects. doi:10.52825/ocp.v5i.1160

RO-Crate Community. (2025). RO-Crate specification 1.2.

The items above are directly aligned to the assessor's critique: they introduce dated, verifiable 2023–2026 benchmark evidence; they explicitly integrate AI as a driver of documentation/knowledge management change; and they strengthen the “user of the materials” perspective with contemporary empirical literature.

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1 25 Bridging the Data Discovery Gap: User-Centric Recommendations for Research Data Repositories
<https://datascience.codata.org/articles/2045/files/698d9ebaf0ef5.pdf>

2 5 8 15 29 Survey Report: Strategic Approaches to and Research ...
https://www.scienceeurope.org/media/jcvjcnpe/202410-survey-report-strategic-approaches-and-research-assessment-open-science.pdf?utm_source=chatgpt.com

3 23 The landscape of data reuse in interactive information retrieval: Motivations, sources, and evaluation of reusability
<https://ideas.repec.org/a/bla/jinfst/v76y2025i9p1258-1276.html>

4 26 28 From Data Creator to Data Reuser: Distance Matters · Issue 7.2, Spring 2025
<https://hdsr.mitpress.mit.edu/pub/2mvqwgmf>

6 Research data management - NWO
https://www.nwo.nl/en/research-data-management?utm_source=chatgpt.com

7 10 Open Science NL Work programme 2024-2025
https://zenodo.org/records/10074873/files/Work%20programme%202024-2025_final.pdf?download=1&utm_source=chatgpt.com

9 27 RESPONSIBLE USE OF GENERATIVE AI IN RESEARCH ...

https://research-and-innovation.ec.europa.eu/document/download/2b6cf7e5-36ac-41cb-aab5-0d32050143dc_en?filename=ec_rtd_ai-guidelines.pdf&utm_source=chatgpt.com

11 16 Guidance for generative AI in education and research

https://unesdoc.unesco.org/ark:/48223/pf0000386693?utm_source=chatgpt.com

12 Artificial intelligence in knowledge management: Identifying and addressing the key implementation challenges - ScienceDirect

<https://www.sciencedirect.com/science/article/pii/S0040162525002148>

13 The Impact of Modern AI in Metadata Management | Human-Centric Intelligent Systems | Springer Nature Link

<https://link.springer.com/article/10.1007/s44230-025-00106-5>

14 18 A Novel Heuristic for Disclosing Uses of AI

https://journals.sagepub.com/doi/full/10.1177/16094069251404329?utm_source=chatgpt.com

17 Authorship and AI tools

https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools?utm_source=chatgpt.com

19 DataCite Metadata Schema 4.5

https://schema.datacite.org/meta/kernel-4.5/?utm_source=chatgpt.com

20 The EOSC Partnership Monitoring Framework

https://eosc.eu/wp-content/uploads/2024/08/20240621_Draft_EOSC_Monitoring_Framework_v7_Final_Clean.pdf?utm_source=chatgpt.com

21 Releases · ResearchObject/ro-crate

https://github.com/ResearchObject/ro-crate/releases?utm_source=chatgpt.com

22 Turning Machine-Actionable DMPs Into FAIR Digital Objects

https://www.tib-op.org/ojs/index.php/ocp/article/view/1160?utm_source=chatgpt.com

24 An empirical examination of data reuser trust in a digital ...

https://www.oclc.org/research/publications/2024/empirical-examination-data-reuser-trust-digital-repository-JASIST.html?utm_source=chatgpt.com